THE UNIVERSITY OF BRITISH COLUMBIA
Faculty of Forestry

Dear Dr. Surridge,

Please consider our paper, entitled "Biological and climatic constraints drive reproductive synchrony" for publication as an Article in *Nature Plants*. Here we answer a fundamental question in plant reproductive biology with relevance to climate change forecasting: what drives synchrony in seed production?

Understanding how plants time reproduction in fluctuating and unpredictable environments is a major goal of plant biology. This timing shapes not only individual reproductive success, but also the reproductive dynamics of whole populations and ecosystems. Across temperate and tropical forests, many species align their cycles so that most individuals reproduce together, driving seed crops that vary dramatically from year to year—a synchronous phenomenon called masting.

What sustains seed production synchrony is currently unknown. Decades of research has focused on climate, but no climatic cue alone is sufficiently regular to explain the observed cycles[1,2]. This suggests the missing mechanism lies not in the environment but within individual tree processes. Building on extensive research on tree reproductive biology, we integrate a well-established reproductive constraint from plant and crop biology: the temporal overlap between fruit maturation and floral bud initiation.

Leveraging 45 years of data, we show how climate and reproductive constraints together produce the classic 'boom' and 'bust' cycles of masting and determine synchrony. We show that cold summers act as a critical hinge, resetting all trees to a common reproductive state and driving long-term synchrony. In contrast to recent studies[3,4] we find no evidence of shifts in masting, and show how reproductive constraints prevent runaway responses to warming. Our results highlight that any shifts in masting would come not from increased seed production, but may come from another mechanism. We forecast decline in synchrony if intermittent cold summers decline—a result with increasing relevance given the extreme rate of recent European summer warming[5].

Our results advance fundamental plant biology by showing how the interplay between individual constraints and shared climatic cues provides a simple physiological mechanism for synchrony, with implications well beyond masting. Our model provides a new and much improved way to forecast how forests will regenerate as the climate warms. Beyond forests, we show how biological constraints can fundamentally limit the runaway amplification of climate effects at larger scales.

All authors contributed to this work and approve this version for submission. The manuscript is 2560 words, with 51 references and 5 figures, and is not under consideration elsewhere. We hope you find it suitable for publication in *Nature Plants*, and look forward to hearing from you.

Sincerely,



Victor Van der Meersch, PhD
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University of British Columbia

Potential reviewers

- Valentin Journé – Kyushu University, Japan (journe.valentin@gmail.com)
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References

[1] J. M. LaMontagne et al. *Nature Plants* (2020).

[2] V. Journé et al. *Nature Plants* (2024).

[3] M. Bogdziewicz et al. *Nature Plants* (2020).

[4] J. J. Foest et al. *Global Change Biology* (2024).

[5] D. L. Schumacher et al. *Communications Earth & Environment* (2024).